

ALLIANCE UNIVERSITY

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NAAC GRADE A+ ACCREDITED UNIVERSITY

Alliance School of Advanced Computing

Bachelor of Computer Applications

ACADEMIC YEAR: 2024-26

Semester – V

Business Analytics

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MINI PROJECT

MINI PROJECT ANSWER BOOK

Power BI Sales Dashboards

For faculty reference and evaluation only

1. Purpose of this answer book

This book gives the expected solution for each of the six mini projects: the DAX measures, the visuals and the verified figures that a correct dashboard should show. Use it to check student submissions quickly. Designs may differ, but the numbers must match.

2. Projects covered

#	Industry	Table name in Power BI	Main value column
1	Real Estate	RealEstate	Sale_Price_INR

3. Common workflow every student should follow

- **Load:** import the CSV or Excel sheet with Get Data. Set Date to the Date type and every INR, quantity and distance column to Whole or Decimal Number.
- **Clean:** check blanks, duplicates and spelling in Power Query. The datasets are clean, so students should show that they checked.
- **Model:** add calculated columns (Month, MonthNo, Quarter) and at least five DAX measures.
- **Visualize:** build KPI cards, a trend line, a category bar, a donut or treemap, a table and at least two slicers.
- **Interpret:** write three to five insights on the dashboard in a text box.

4. Expected dashboard layout

Area	Content
Top strip	Dashboard title with industry name, plus 3 KPI cards
Left panel	Slicers: Date, main category, one more dimension
Middle left	Line chart of sales by month
Middle right	Bar chart of sales by main category
Bottom left	Donut or treemap of the second dimension
Bottom right	Detail table and text box with insights

Note: *all figures in this book were calculated from the datasets issued to students. Amounts are in INR; Cr = crore (10,000,000) and L = lakh (100,000).*

Project 1: Real Estate Sales Dashboard

Dataset columns: Deal_ID, Date, Property_Type, City, Area_sqft, Agent, Sale_Price_INR

A. Key DAX measures (Table: RealEstate)

Measure name	DAX formula
Total Sales Value	SUM(RealEstate[Sale_Price_INR])
Total Area Sold (sqft)	SUM(RealEstate[Area_sqft])
Deals Closed	COUNTROWS(RealEstate)
Avg Price per sqft	DIVIDE([Total Sales Value], [Total Area Sold (sqft)])
Top City	MAXX(TOPN(1, VALUES(RealEstate[City]), [Total Sales Value]), RealEstate[City])
Premium deals (above ₹2 Cr)	CALCULATE([Deals Closed], RealEstate[Sale_Price_INR] > 20000000)

B. Expected visuals

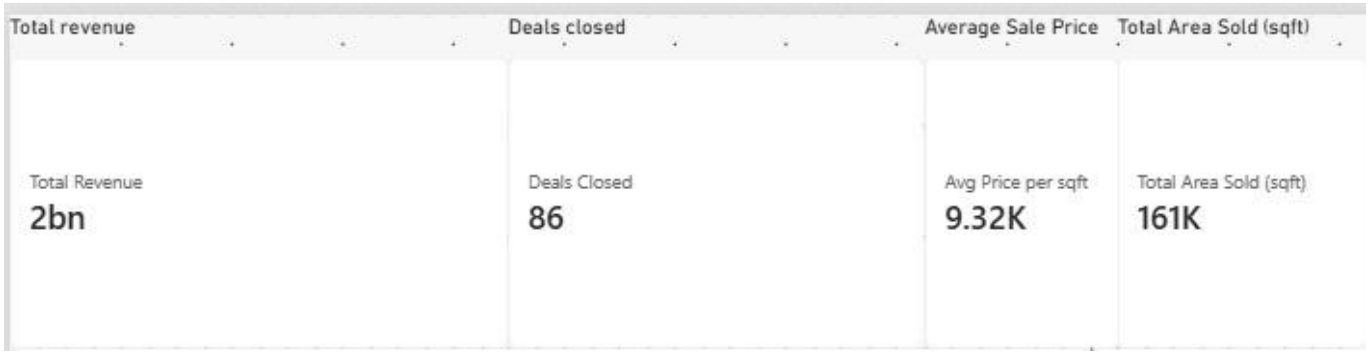
Visual	Fields / settings
KPI cards	Total Sales Value, Deals Closed, Avg Price per sqft
Line chart (trend)	Axis: Month (sort by MonthNo) | Values: Total Sales Value
Clustered bar	Axis: Property_Type | Values: Total Sales Value (sort descending)
Donut / Treemap	Legend: City | Values: Total Sales Value
Table	Deal_ID, Date, Property_Type, City, Agent, Sale_Price_INR
Slicers	Date, Property_Type, City, Agent

C. Answer key: expected values and insights

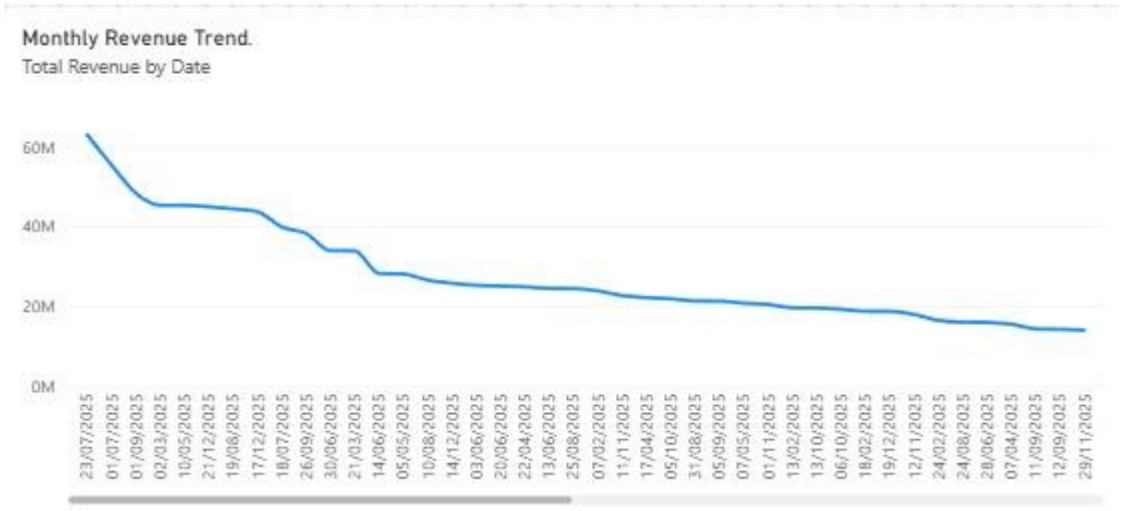
- **Check values:** Total Sales Value = ₹174.56 Cr; Deals Closed = 100; average per record = ₹1.75 Cr; premium deals (above ₹2 cr) = 33.
- **Insight 1:** Villa is the top property type with ₹53.52 Cr (30.7% of total); Studio is the lowest at ₹6.65 Cr.
- **Insight 2:** Mumbai leads by city (₹37.76 Cr, 21.6%). The best month is December (₹23.52 Cr) and the weakest is January (₹6.75 Cr).
- and an insights text box.

D. Post the Dashboard Screenshots here:

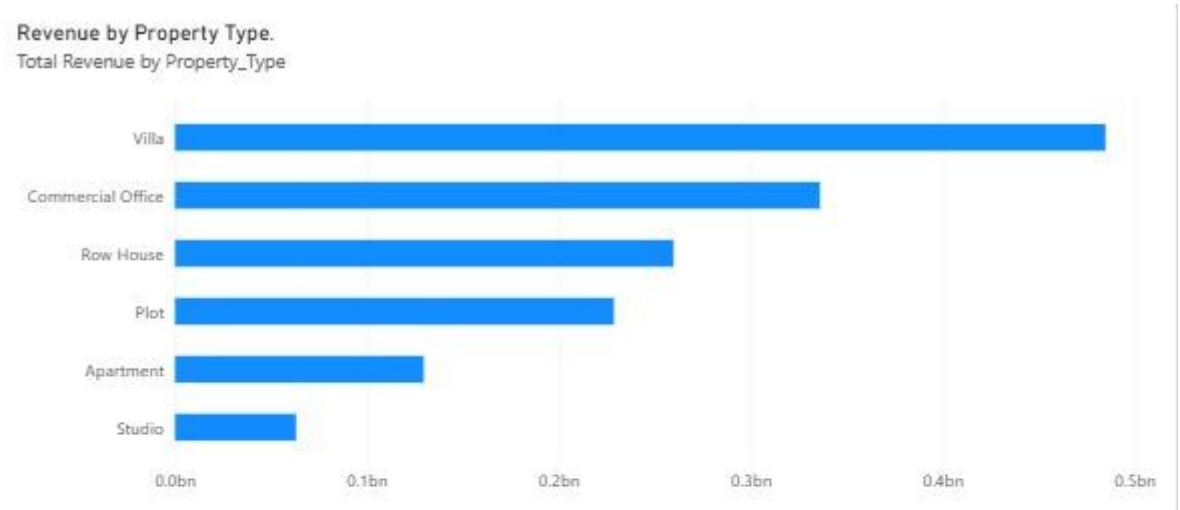
1. KPI cards

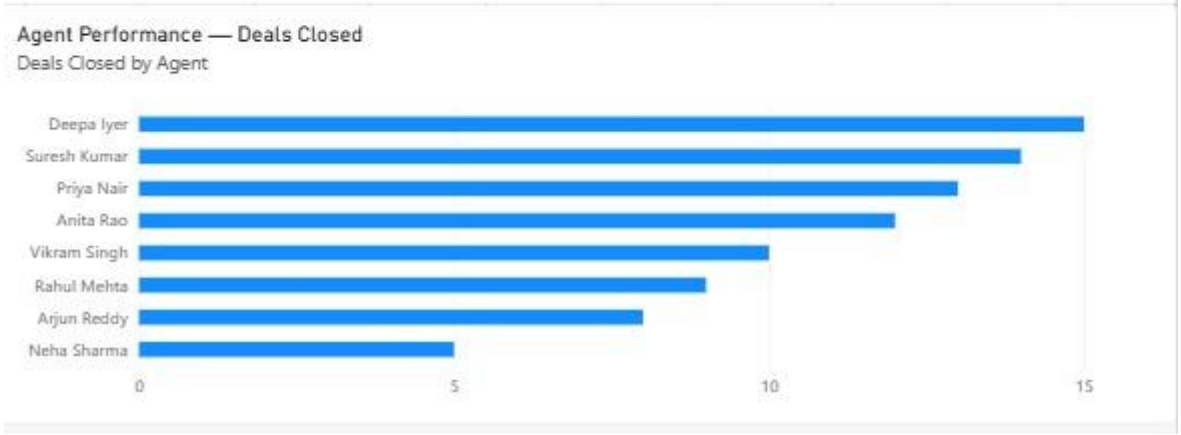


2. Line Chart

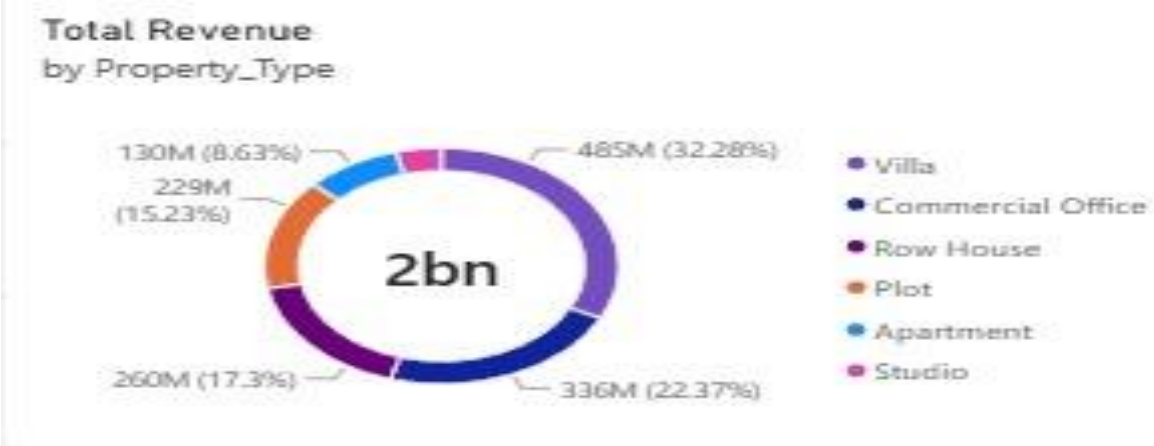


3. Clustered Bar chart





4. Donut chart



5. Table

Deal_ID	Date	Property_Type	City	Area_sqft	Agent	Sale_Price_INR
RE5001	26/10/2025	Apartment	Hyderabad	1990	Rahul Mehta	13600000
RE5002	21/08/2025	Studio	Mumbai	460	Deepa Iyer	4160000
RE5003	25/08/2025	Commercial Office	Chennai	2810	Neha Sharma	24430000
RE5004	02/01/2025	Row House	Chennai	1240	Suresh Kumar	10880000
RE5005	26/06/2025	Studio	Chennai	370	Rahul Mehta	1930000
RE5006	05/02/2025	Studio	Hyderabad	360	Anita Rao	3480000
RE5007	11/03/2025	Plot	Chennai	2870	Deepa Iyer	10020000
RE5008	13/07/2025	Plot	Hyderabad	2770	Rahul Mehta	13760000
RE5009	28/06/2025	Row House	Bengaluru	2300	Priya Nair	15940000
RE5010	25/04/2025	Studio	Bengaluru	450	Anita Rao	3440000
RE5011	06/10/2025	Plot	Hyderabad	3100	Deepa Iyer	19230000
RE5012	21/03/2025	Villa	Delhi NCR	2810	Suresh Kumar	33910000
RE5013	05/12/2025	Row House	Hyderabad	1430	Anita Rao	8900000
RE5014	01/11/2025	Villa	Bengaluru	2640	Arjun Reddy	20430000
RE5015	25/01/2025	Studio	Mumbai	520	Anita Rao	4430000
RE5016	22/04/2025	Commercial Office	Bengaluru	2300	Priya Nair	24870000
RE5017	22/10/2025	Studio	Delhi NCR	420	Vikram Singh	2590000
RE5019	13/10/2025	Commercial Office	Pune	1730	Priya Nair	19550000
RE5021	29/07/2025	Studio	Mumbai	430	Neha Sharma	3650000
RE5022	31/08/2025	Commercial Office	Mumbai	1250	Suresh Kumar	21320000
RE5023	30/01/2025	Plot	Pune	1790	Deepa Iyer	12410000
RE5024	18/07/2025	Commercial Office	Mumbai	2780	Neha Sharma	39920000
RE5025	19/08/2025	Villa	Delhi NCR	3150	Rahul Mehta	44460000

E. Conclusion:

Conclusion

The Real Estate Sales Dashboard was successfully developed using Microsoft Power BI to analyze and visualize real estate sales data. The project involved data cleaning and transformation using Power Query, creating DAX measures, and developing interactive visualizations to present meaningful business insights.

The dashboard highlights key performance indicators (KPIs), including total sales value, total area sold, number of deals closed, and average price per square foot. Visualizations such as monthly revenue trends, revenue by property type, city-wise revenue, agent performance, and a donut chart help identify sales patterns and compare business performance. Interactive slicers enable users to filter data based on city, property type, and date for more detailed analysis.

Overall, this project demonstrates how Microsoft Power BI can transform raw data into meaningful and actionable insights. The dashboard helps managers evaluate sales performance, identify high-performing cities and property types, assess agent performance, and make informed, data-driven decisions. Furthermore, this mini-project enhanced practical knowledge of data analysis, data visualization, DAX calculations, data transformation, and interactive dashboard development.